

Devdoots Assignment 06: Core Python & OOPs Fundamentals Drill (Zero-AI Protocol)

Team: Devdoots | **Team Leader:** Divyansh

Target: Strengthening Core Logic (Lists, Tuples, Strings, Dicts, Loops, Functions & OOPs)

Submission Deadline: Tuesday, August 11, 2026, before 9:00 PM IST



STRICT ZERO-AI PROTOCOL & MANDATORY CODE DEFENSE

Is assignment ka maksad AI se code likhwana bilkul nahi hai. Is baar **ChatGPT, Gemini, Copilot, ya kisi bhi AI tool ka use strict PROHIBITED** hai.

MANDATORY RULE: GitHub par Pull Request (PR) raise karne ke baad, **har team member ko apna poora code Line-by-Line explain karna hoga**. Agar koi member apni kisi line ka logic explain nahi kar paaya, toh submission reject maani jayegi.

1. Prerequisite Study Guide (Kya Revise Karna Hai?)

Is assignment ko khud se attempt karne se pehle ye basic topics revise karein:

- **Strings & Manipulations:** `.lower()`, `.split()`, indexing, slicing, concatenation.
- **Lists & Tuples:** List iteration using for loop, tuple unpacking, `.append()`.
- **Dictionaries:** Key-Value pairs access, `dict[key] = value`, checking key existence with `if key in dict`.
- **Control Flow & Loops:** for loop, if-elif-else conditions, manual logic building.
- **Functions & OOPs Basics:** Function parameters, return statement, class, `__init__`, `self`, and object creation.

2. Industry Relevance (Kyon aur Kahan Kaam Aayega?)

- **Coding Interview Assessments:** Top tech companies aur AI startups ke live coding rounds mein AI allowed nahi hota. Wahan basic Data Structures (List/Dict/String) par apka apna hand-on logic test hota hai.
- **Real Data Cleaning & NLP Pipelines:** Text Datasets (LLM training data) ko clean karna, word frequency count karna, aur custom data points handle karne ke liye basic Python loops aur dictionaries hi core engine hoti hain.

3. Assignment Tasks (Easy to Progressive - Build Logic Step-by-Step)

Task 1: Text Frequency Counter (String & Dict Logic)

Requirement: Ek function `word_counter(text)` banayein jo di gayi string se har word ki frequency (kitni baar aaya) count kare.

- Pehle poori string ko `.lower()` karke `.split()` se words ki list banayein.
- Ek empty dictionary `freq = {}` banayein.
- for loop chalakar har word ko dict mein count karein.
- **Example Input:** "Python is easy and Python is powerful"
- **Expected Output:** {'python': 2, 'is': 2, 'easy': 1, 'and': 1, 'powerful': 1}

Task 2: Product Filter & Manual Max Search (List & Tuple Logic)

Requirement: Ek script likhein jo products ke data par loop chalakar filter kare aur sabse mehanga product dhundhe.

- Products list format: `products = [("Laptop", 50000, 4.5), ("Mouse", 500, 3.8), ("Keyboard", 1500, 4.2), ("Monitor", 12000, 4.7)]` (Format: (name, price, rating)).
- **Part A:** Loop chalakar un sabhi products ka naam print karein jinki rating **4.0 ya usse zyada** hai.
- **Part B:** Inbuilt `max()` function ka use KIYE BINA, for loop aur if condition se sabse mehanga product (Highest Price) dhundhein aur uska naam/price print karein.

Task 3: Basic Student Management Class (OOPs Basics)

Requirement: Ek simple Student class banayein bina kisi complex library ke.

- `__init__(self, name, roll_no, marks):` Student ki details initialize karein. (Note: marks ek dictionary hogi, e.g., {"Python": 85, "Maths": 90}).
- Method `get_average_marks(self):` Dictionary ke marks ka sum karke average calculate kare aur return kare.
- Method `display_details(self):` Student ka naam, roll number, aur average marks clean format mein print kare.

4. GitHub Submission Rules & Workflow (Single File Push)

Standard TPM Single-File workflow follow karein:

Step	Git Command	Description
1. Sync Main	<code>git checkout main && git pull origin main</code>	Latest changes sync karein.
2. Create Branch	<code>git checkout -b user/yourname-assign6</code>	Apni distinct branch banayein.
3. Create File	File: <code>yourname_assign6.py</code>	Sirf apni single file banayein.

Step	Git Command	Description
4. Stage & Commit	git add yourname_assign6.py git commit -m "[HW-6] Core Python Basics Practice"	Direct git add . mat karna.
5. Push & PR	git push origin user/yourname-assign6	GitHub par PR create karein.

5. Post-Submission Viva & Line-by-Line Explanation Process

- PR raise hone ke baad, team meeting mein har member apni Screen Share karega.
- Leader/Peers code ki kisi bhi line ko point karke poochnge: *"Is line par variable me kya store ho raha hai aur loop yahan kaise chal raha hai?"*
- Un-aided explanation complete hone par hi PR merge ki jayegi.